



ADAPT: An Agentic AI Framework for People with Disabilities and Neurodivergence

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ABSTRACT

The present paper introduces ADAPT, an agentic artificial intelligence (AI) framework aimed at assisting people with disabilities and neurodivergent conditions to adopt healthier eating habits and organized daily life. It includes four specialized agents (meal planning, adaptive reminders, food guidance, and physiological monitoring), a Large Language Model (LLM)-based decision layer, a Model Context Protocol (MCP) routing layer, and a multimodal interface. Each agent is deployed in a Perception–Reasoning–Action (PRA) loop and communicates with each other using a shared blackboard and a hybrid reasoning component that uses rules and reinforcement learning. In addition to electronic health records (EHRs), nutritional databases, wearable and Internet of Things (IoT) devices, and smart kitchen gadgets, the framework includes modules for explainable AI (XAI) to provide understandable explanations to the user and the caregiver. A concrete use case is presented showing how the implemented prototype can help a neurodivergent person with diabetes by interpreting multimodal inputs, evaluating meal suitability, making real-time replacements, and providing sensory-sensitive prompts. Results of a synthetic simulation-based evaluation (500 simulated profiles, no real user evaluation) show a positive impact on nutritional adherence and reminder responsiveness, and usability was found to be appropriate across various accessibility profiles. The envisioned framework provides a more unified approach to assistive health technologies and a future path toward personalized, safe, and transparent digital health assistance for individuals with disabilities and neurodivergence.

KEYWORDS

Agentic AI, Disabilities, Neurodivergence, Healthy eating, Assistive technology, Personalized healthcare, Meal planning, Adaptive reminders

INTRODUCTION

Daily routines and a healthy diet are widely recognized as important contributors to human wellbeing. A balanced diet, frequent reminders, and positive counseling may support long-term quality of life, psychological stability, and physical health (World Health Organization, World Bank, 2011). Nevertheless, keeping such habits poses some specific challenges to neurodivergent people and people with disabilities. Mobility issues, sensory issues, executive-functioning issues, and medical issues requiring a special diet may contribute to difficulties. These populations often have inadequate access to traditional health applications and digital wellness platforms because of their inability to be personalized, inclusive, and adaptive to their needs and interests, which is why these applications are ineffective in the given scenario (UNICEF, 2021).

The cost of disability is high at the international level. According to the World Health Organization (WHO), >1 billion or approximately 15% of the global population lives with a disability (World Health Organization, World Bank, 2011).

Neurodivergent individuals are another category of population that is added to this group, including dyslexia, attention deficit hyperactivity disorder (ADHD), autism spectrum disorder (ASD), and other cognitive impairments. These people typically struggle to obtain the right kind of healthcare system, nutrition, and assistance related to lifestyle (Kapp *et al.*, 2013). Popular digital health solutions provide general recommendations, but do not take the complexity of interaction between cognitive, behavioral, and physiological needs into account, which constitute neurodiversity and disability.

Recent advances in artificial intelligence (AI) now make more inclusive healthcare and lifestyle support a possibility. Personalized meal planning systems based on machine learning have been demonstrated to customize diets to some medical conditions such as diabetes and hypertension. The subject-oriented voice recognition system and the AI vocabulary expansion are also essential to support the speech patterns of older users in the communication process and enlarge

the scope of their vocabulary use in communication with AI-powered devices, which must be extended to facilitate their use in communication (Kiran *et al.* 2024); (Hass, 2020). Wearable and mobile interfaces to provide adaptive reminders have been shown to be effective in assisting individuals with cognitive impairments in their daily task management. Likewise, food guidance and monitoring applications have been utilizing Internet of Things (IoT) sensors, image recognition, and natural language processing (NLP) to monitor what has been consumed and offer real-time feedback on the same. Most of these systems, however, lack a cohesive, holistic approach, as most of them concentrate on only one aspect of healthy living.

Agentic AI is a promising shift in paradigm. Orchestrating autonomy, reasoning, and proactive decision-making, and agentic AI stands in contrast to regular rule-based or passive recommendation systems. These systems consist of a number of specialized agents that interoperate dynamically to satisfy the evolving needs of a user, for example, an intake monitor, a behavioral coach, and a dietary planner. Agentic AI has already shown promise in the healthcare sector for managing chronic illness conditions, medication non-adherence, and patient telemonitoring. Older adults with difficulty in malnutrition (low protein intake because of chewing problems and impaired sense of taste) are also encouraged to use AI-powered robots that operate on machine learning algorithms to create healthy eating habits based on their preferences and needs (Jan *et al.*, 2025a,b; Kalu *et al.*, 2025). Its possible contribution to the benefit of individuals with neurodivergence and disabilities is untapped.

To address the urgent need for genuinely personalized and adaptive digital health interventions for this underserved population, we propose ADAPT: a closed-loop multi-agent AI design specifically architected for the disability and neurodivergence context. Our system, as explained in Figure 1, combines user communication (voice, text) and

advanced intent interpretation, which is powered by a Large Language Model (LLM). This will enable dynamic routing of tasks to special modules including the Meal Planner and Reminder agents. The main element of this system is the Central Reasoning Engine that integrates codified rules with reinforcement learning (RL) to combine real-time inputs from the Data Integration Layer [electronic health records (EHRs) and IoT] with the Central Reasoning Engine. This learning cycle will be continuous, which means that the Recommendations & Guidance that are shown to the user and caregivers are not fixed, but may change depending on the specific case and outcome, creating the basis of next-generation personalized care.

This paper makes three targeted contributions. First, it offers a comprehensive literature review that highlights gaps in the field of providing dietary support with AI, assistive technologies, and multi-agent health systems in relation to disability and neurodivergence. Second, it suggests a coherent model for agentic AI, which encompasses all aspects of meal planning, adaptive reminders, food guidance, and monitoring in a single model for this population combination, which has not been assembled in the literature so far. The novelty of the central system is that the system is built with the components that have been used before to build agent-based and LLM-based systems, but they are made accessible-by-design for users with disabilities and other neurodivergent users, and they are connected via Model Context Protocol (MCP) as a disciplined routing protocol from the language-level reasoning to the specialized agents, and these agents are applied to a population that has been largely ignored in the previous multi-agent health systems. Third, the paper provides a prototype implementation and simulation-based evaluation that shows the feasibility of the framework and its capacity to enhance adherence and decrease caregiver burden.

This is the organization of the rest of the paper. In the Background section, the background information on

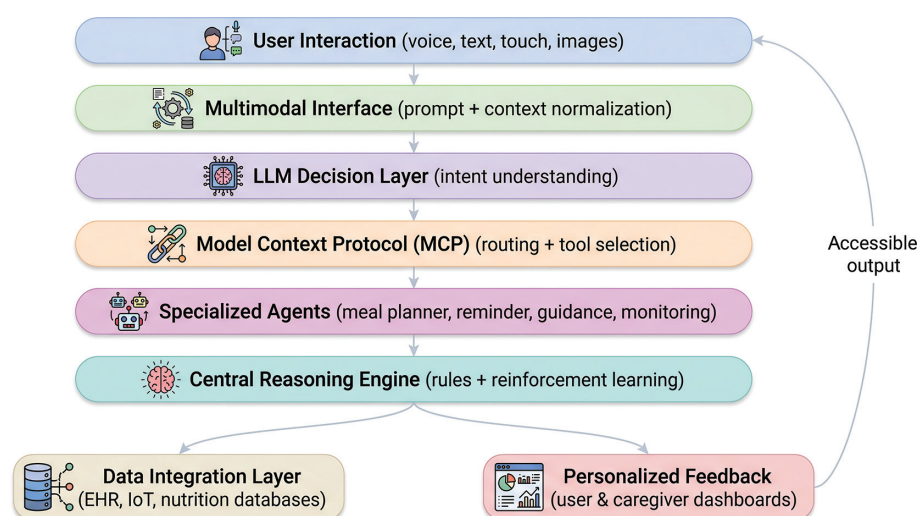


Figure 1: Architecture of an AI-driven health and wellness system. A closed-loop pipeline processes user input through a multimodal interface and LLM for intent understanding, routes tasks via MCP to specialized agents, and integrates outputs in a central reasoning engine (rules + reinforcement learning). Context from EHR and IoT data enables personalized recommendations and feedback. Abbreviations: AI, artificial intelligence; EHR, electronic health record; IoT, Internet of Things; LLM, Large Language Model; MCP, Model Context Protocol.

neurodivergence, disabilities, and the role of structured routines in health management is provided. The Literature Review section gives a detailed literature review of the work that has been done in dietary recommendation systems, assistive technologies, and multi-agent healthcare applications in the past. The proposed agentic AI architecture will be discussed in the Proposed Agentic AI Framework section and its implementation and preliminary results in the Implementation and Results section. The Limitations section provides the key limitations of the study. The Conclusion section provides conclusion and future directions of the paper.

A preliminary conceptual outline of this framework was disseminated as an arXiv preprint (Jan *et al.*, 2025b) during the manuscript development period. That preprint was merely a conceptual introduction to the four-agent structure and the Perception–Reasoning–Action (PRA) loop. The present journal submission offers a substantial extension of that preliminary account in the following ways: (1) 500 synthetic profiles of users were evaluated, across 8 weeks, with a full simulation-based evaluation presented, which provides concrete performance metrics for the four agents presented (nutritional adequacy, adherence, usability, caregiver burden reduction); (2) four complete algorithmic specifications are provided, each with structured formal pseudo-code for the Meal Planner (Q-learning), the Reminder (contextual bandits), the Food Guidance [convolutional neural network (CNN)/BERT multimodal classification], and the Monitoring [Gated Recurrent Unit (GRU)-based anomaly detection] agents; (3) a detailed safety, privacy, and governance framework is formalized, including regulatory pathway analysis under criteria of the FDA and EU MDR 2017, data-minimization principles, and explicit clinical edge-case handling logic as structured text; (4) the MCP routing architecture and external connector protocol are fully specified; (5) an accessibility analysis is presented by disability condition (six distinct population subgroups); and (6) a comprehensive comparative literature review is presented, with a multi-dimensional comparison table (Table 1) also included. This relation is related to the prior conceptual account of the preprint and disclosed to the editorial team as per the policy of the journal.

BACKGROUND

Neurodiversity, disability, and health management are complex and multifaceted challenges when they are combined. In order to encourage the development of the agentic AI systems that will be able to address the specific challenges faced by these groups, it is essential to understand this context.

Disabilities and health challenges

According to the WHO, the term disability is an umbrella term that entails restrictions on participation, activity

Table 1: Comparative analysis of ADAPT against prior multi-agent and AI health frameworks.

System	Target population	Multi-agent	Accessibility design	EHR/IoT	Meal planning	XAI	Key limitation
ADAPT (this work)	Disability/neurodivergence	Yes (4 agents + MCP)	Yes (core principle)	Yes	Yes (RL-based)	Yes	Simulation-only evaluation; no real user study yet
Papastratis <i>et al.</i> (2024)	General clinical users	No	No	Partial	Yes (deep generative)	No	Single model; no agents; no accessibility
Guo <i>et al.</i> (2025)	General (dietary adherence)	No	No	No	Yes (LLM-based)	No	No disability focus; no agents; short-term adherence only
Syed <i>et al.</i> (2026a,b)	General (budget-constrained users)	Yes (agentic)	No	No	Yes (budget-aware nutrition)	No	No disability/accessibility design; budget-centric scope
Kiran <i>et al.</i> (2024) and Hass (2020)	Elderly type II diabetes patients	No	Partial (elderly, nudging UI)	No	Yes (food recommendation)	No	Single-domain (diabetes only); no agents; no neurodivergence support
Esteve <i>et al.</i> (2026)	Elderly users	No	Partial (voice, elderly-focused)	No	Partial (cooking guidance only)	No	Cooking-only scope; no disability/neurodivergence; no multi-agent; no EHR/IoT
Mohammed, Ibrahim, and Cavus (2018)	Older adults with disabilities	No	Yes (disability-targeted)	No	No	No	Medication reminders only; no diet planning; no multi-agent; no EHR/IoT

Entries for prior systems are based on information reported in the cited publications.

Abbreviations: Accessibility design, accessibility for sensory, motor, and cognitive differences as a core architectural principle; EHR, electronic health record; IoT, Internet of Things; LLM, Large Language Model; MCP, Model Context Protocol; RL, reinforcement learning; UI, user interface; XAI, Explainable AI.

limitation, and impairment (World Health Organization, World Bank, 2011). Abnormalities in physical, sensory, cognitive, and psychosocial ability often accompany long-term medical conditions, which demand careful lifestyle adjustment. As an example, victims of spinal cord injury may lack the ability to move around, and therefore, cooking, shopping, and preparing healthy dishes will be hard. Although intellectually disabled persons may struggle to comprehend dietary guidelines or manage the amount of food to eat, the visually impaired persons may struggle to access nutrition data. Such limitations make conventional health management tools less effective and available.

Nutrition is essential in order to alleviate secondary health conditions that are associated with disability. AI-based approaches have demonstrated particular promises in diabetes management, spanning glucose monitoring, personalized dietary planning, and real-time adherence support (Guan *et al.*, 2023). People with disabilities face a disproportionate risk of malnutrition, obesity, and cardiovascular conditions compared to the general population (Krahn *et al.*, 2015—a finding that remains broadly consistent with more recent epidemiological data). As per the latest perspective, the environments of health services are the least integrated with AI technology (Garg *et al.*, 2018; Alowais *et al.*, 2023; Aravazhi *et al.*, 2025). Social stigma, lack of access to healthcare, and reliance on caretakers may exacerbate these disparities. Interventions to facilitate self-management and healthy eating are needed in order to enhance long-term outcomes.

Neurodivergence and lifestyle routines

Cognitive dissonances that do not reflect the standard are known as neurodivergence (Kapp *et al.*, 2013). Some of these are ASD, ADHD, dyslexia, dyspraxia, and Tourette syndrome. Neurodivergent people often have problems with behavioral control, executive functions, and sensory integration. These challenges directly affect meal planning, grocery shopping, food preparation, and meal-eating schedules (Jan *et al.*, 2025a,b).

To illustrate, food preferences could be affected by the increased sensory sensitivity in autistic people, which leads to restrictive diets that are not balanced in nutritional content (Bandini *et al.*, 2017). AI is very useful in nutritional analysis. It requires minimal user input and can measure and analyze foods through image recognition. The latest machine learning methods can deliver nutrient profiles instantly by analyzing photos of meals, which is achievable by the latest machine learning techniques that analyze the photos of meals in real time, without the need to know the food items involved in the meals (Zheng *et al.*, 2024). The ADHD sufferers could experience difficulties with their time management and impulsivity, and this may result in unstable dietary habits or reliance on convenience foods (Faraone *et al.*, 2021). These lifestyle pitfalls can worsen such long-term health issues as obesity, diabetes, and gastrointestinal disorders. Structured routines, visual cues, and adaptive systems are important in order to assist neurodivergent populations to live healthier lifestyles.

Importance of structured routines

Healthy routines provide predictability and stability, which can be particularly beneficial to people with neurodivergence and disabilities. Routine activities assist in adherence to medications, sleep–wake cycles, anxiety reduction, and energy expenditure management. Especially important in the maintenance of metabolic fitness are nutritional habits such as portioning and eating schedules. The reality is, however, that such routines may be difficult to keep without external help. Technology-assisted interventions such as reminder applications and calendar notifications have also shown potential in enhancing the level of adherence. Systematic reviews of mHealth interventions for chronic disease management have confirmed measurable improvements in treatment adherence and patient outcomes (Hamine *et al.*, 2015).

However, these solutions are often not customized and flexible. They do not consider the variation in cognitive abilities of people, sensory preferences, and medical needs since they predict the same user behavior. As a paradigm centered on individual user-specific patterns, the dynamic adaptation of interventions, and proactive assistance (as compared to a passive notification), agentic AI offers a possibility of bridging this gap.

Current digital health limitations

The availability of wellness coaching, exercise tracking, and food tracking has been augmented by the proliferation of mobile health (mHealth) applications. Although there are more specific applications that help in the management of chronic illnesses such as diabetes, other applications such as MyFitnessPal and Noom are popular and have behavioral nudges and calorie-tracking features. Although these platforms are common and often used, they are often inaccessible to people with disabilities due to various factors such as non-inclusive content, overpowering interfaces, or a lack of compatibility with screen readers. In addition, neurodivergent users tend to complain about the rigid and prescriptive systems that fail to accommodate their needs in terms of sensory and mental capabilities of an individual user misfit in most aspects.

Siloed functionality is another weakness. These current tools might be limited to tracking meals, reminding you to eat, or recommending recipes, but they do not necessarily work in a unified package to support a lifestyle. Clinicians and caregivers are frequently required to put together various data that may not be related to each other, resulting in a lack of services provided and poor health outcomes. This highlights the need for unified structures that are capable of making smart, autonomous judgments and are accommodative in nature.

Agentic AI as an opportunity

The ability to act autonomously, collaborate with other agents, and work toward user-oriented goals in dynamically changing environments differentiates agentic AI, which

is grounded on multi-agent system (MAS) architectures. Unlike the non-agentic recommender systems, agentic AI adjusts to changes in user preferences, context, and environment. These systems are managed by a central reasoning engine and may be specialized agents for specific tasks in the healthcare sector, like behavior monitoring, nutritional analysis, or adaptive scheduling. Syed *et al.* (2025, 2026a,b) proposed frameworks for integrating finance-aware and nutrition-aware agentic AI systems. Due to these features, agentic AI is particularly the most appropriate to tackle the problem of neurodiversity and disability, where accessibility, flexibility, and personalization are essential factors to consider.

It has been demonstrated that technological advancements can expose opportunities that could potentially provide care to the elderly in a specialized way and manner of their own preference, as well as provide food and beverages, AI, and robotics (Kramarow, Warner and Chen, 2014; Salzmänn-Erikson, 2023; Rashid and Kausik, 2024). Designing agentic AI systems can transform digital health to serve vulnerable populations and include inclusivity, accessibility, and transparency in its design. They may assist individuals to be more independent, reduce the burden of taking care of other people, and promote healthier lifestyles, which may result in improved quality of life and more social interactions.

LITERATURE REVIEW

Extensive research has been done on digital interventions for diet, health management, and assistive technologies. Viewed through the prism of agentic AI, however, there are few studies that explicitly combine these with neurodivergent and disabled individuals. The six themes discussed in this section, in terms of the relevant literature review, include dietary recommender systems, disability aid technologies, AI and neurodivergence, multiple-agent healthcare, adaptive reminders and behavioral nudges, and intake monitoring through the use of IoT/AI. It is on the basis of these works that we have proposed an agentic AI framework.

Dietary recommender systems

Dietary recommender systems have experienced a dramatic change over the last decade due to the evolution of machine learning, personalization algorithms, and human-computer interaction. Specifically, machine learning and deep-learning methods can greatly enhance the accuracy and performance of the diet assessment and observation of eating habits (Salinari *et al.*, 2023; Theodore Armand *et al.*, 2024). Among the earliest approaches to nutrition-conscious recommendation, Freyne and Berkovsky (2010). offered an approach that prioritizes the importance of finding a compromise between user preferences and health limitations, emphasizing the significance of striking a balance between user preferences and health constraints. This was followed by subsequent research that incorporated user profiles, contextual factors, and health goals. A large-scale European

randomized controlled trial demonstrated that personalized nutrition interventions produce significantly greater behavioral changes compared to generalized dietary advice (Celis-Morales *et al.*, 2017). For instance, Guan *et al.* (2023) proposed an RL-based system that recommends a balanced diet to diabetic patients by clustering. Meyers *et al.* (2015) used deep learning to recognize food images, and they incorporated this technique into a smartphone in order to track a diet and give feedback in real-time based on the diet.

Designing health-aware meal recommendation algorithms has been an ongoing pursuit in the field (Freyne and Berkovsky, 2010; Freyne *et al.*, 2011; Harvey, Ludwig and Freyne and Berkovsky (2010); Ge, Ricci and Massimo, 2015; Ge *et al.*, 2015). With the advent of LLMs, the design space has grown significantly more recently. The study conducted by Guo *et al.* (2025) on meal planning with ChatGPT showed that, in controlled environments, dietary interventions using LLM could greatly enhance short-term adherence, but sustainability and depth of personalization are still challenges. Similarly, Papastratis *et al.* (2024) proposed a deep generative model combined with ChatGPT for the AI-based nutrition recommendations, achieving high accuracy in a clinical pilot, and calling for multimodal sensing.

AiNutrify is a related work that used the CrewAI multi-agent framework to build a closed-loop LLM-based meal planning personal assistant that shares concepts with ADAPT's agent-level structure, but does not consider accessibility requirements or the neurodivergent population. Another line of inquiry was the incorporation of medical guidelines into recommendation logic. To make changes in the suggestions dynamically, Papastratis *et al.* (2024) suggested a hybrid tool that would integrate reported symptoms of the user with nutrition databases. Similarly, Ge *et al.* (2015a, 2015b). examined the application of collaborative filtering to the proposal of low-sodium diets to individuals with cardiovascular diseases. These works bring up the feasibility of AI-based personalization but also point to its limitations: most of the systems target specific medical groups, ignore the problem of accessibility, and rarely consider neurocognitive diversity.

Recently, motivational factors of diet adherence have been examined as well. As an example, in comparison with generic guidance, Celis-Morales *et al.* (2017). concluded that customized web-based dietary feedback enhanced adherence significantly. The interface was, however, not user-friendly with individuals who were cognitively challenged or with less digitally literate individuals. This is an important observation: accessibility is equally vital to the process of ensuring equity of results as personalization; by employing proactive modifications, multimodes, and adaptive interfaces, designer agentic AI can close this gap.

Assistive technologies for disabilities

Assistive technological research has been aimed at providing more independence and quality to the lives of people with disabilities alongside the development of dietary recommendation systems. Assistive technologies are as simple as a screen reader and as complex as AI-based systems. According to

Krahn *et al.* (2015) assistive technologies have two functions: they facilitate social inclusion and make up for the functions of the impaired person. This can be voice-enabled kitchen appliances, available recipe websites, and simplified shopping assistants as far as diet plans are concerned.

There are already promising digital health tools that are specifically developed with people with intellectual and developmental disabilities (IDDs). Mohammed *et al.* (2018) demonstrated that the use of mobile reminder apps by adults with intellectual disabilities improved task completion and independence in their assessment. The study did, however, emphasize the need to have adaptive systems that are capable of adjusting complexity levels to match cognitive abilities. On the same note, mainstream nutrition applications are often not designed in line with accessibility standards, thereby restricting their functionality to those with visual or motor disabilities, as demonstrated in prior accessibility studies.

Advanced systems must support numerous input modalities (e.g., voice, gesture, touch) and adjustable narration speeds to make them more usable by individuals with hearing or vision deficiencies (Cheng *et al.*, 2018; Seiderer, Ritschel and André, 2020; Di Martino, Delmastro and Dolciotti, 2021; Kiran *et al.*, 2024; So *et al.*, 2025). Empirical evaluation of voice assistants with impaired users has confirmed that while such interfaces offer meaningful accessibility benefits, unresolved barriers persist for users with complex communication needs (Masina *et al.*, 2020). Recent studies have employed AI in enhancing inclusivity. As an example, Masina *et al.* (2020) recommended a voice-activated helper that integrates the visual impairment accessibility features with dietary notifications. Similarly, Esteve *et al.* (2026) introduced an intelligent cooking assistant that blends voice-delivered instructions with detailed adaptive instructions to support the motor-impaired individuals in cooking. More recent work has explored proactive conversational cooking companions for elderly users as a promising direction in assistive dietary support (Esteve *et al.*, 2026). Our proposed agentic system builds on these pieces and must offer personalization, autonomy, and multi-agent collaboration, besides being accessible.

The importance of co-design is also highlighted in the greater literature of disability and technology. A study by Shinohara and Wobbrock (2016) revealed that including people with disabilities in the design process can lead to more inclusive and successful results. This concept aligns with the philosophy of agentic AI that requires autonomous action in addition to adaptive learning to user feedback in an open and interactive manner.

AI and neurodivergence

The fields of education, work, and healthcare have studied how AI can help neurodivergent people. The authors of the article by Kapp *et al.* (2013) indicate that inclusive technologies must encourage autonomy and identity and self-regulation instead of deficit models. Neurodiversity often influences behavioral control and sensory preferences in the context of diet, which leads to restrictive diets or abnormal

eating habits (Bandini *et al.*, 2017). The personalization of recommendations, based on sensory sensitivity and maintaining nutritional balance, can be used to accommodate these trends with the help of AI.

Many studies have pointed out the potential of AI to aid executive-functioning problems that are characteristic of autism and ADHD. As an example, Faraone *et al.* (2021) discovered that although machine learning models could forecast adherence patterns, the use of digital reminders increased adherence to structured routines in children with ADHD. Similarly, Daniels *et al.* (2021) proposed the use of AI-based behavioral tracking devices to assist individuals with autism to cope with their daily activities. These articles indicate the applicability of adaptive technologies to compensate for the deficiencies in self-monitoring, time management, and planning.

AI has also been used in augmentative and alternative communication (AAC). Neurodivergent groups can become more active and self-reliant using predictive keyboards, text-to-speech applications, or visual schedule applications, etc. (Light *et al.*, 2019). Since this ensures the availability of the population with communication challenges, it is necessary to integrate such multimodal supports in dietary and health management systems. Digital health interventions for children and young people with mental health problems have shown meaningful effects on symptom management, further supporting the case for technology-based approaches across neurodivergent populations (Hollis *et al.*, 2017). Importantly, autistic adults themselves are recognized as critical autism experts whose perspectives should inform the design of inclusive AI tools (Gillespie-Lynch *et al.*, 2017). The latter studies demonstrate that AI is especially flexible and personalized, which is why it suits the needs of neurodivergent people rather well.

Multi-agent healthcare systems

MASs, which are entities that offer autonomy, scalability, and modularity, have increased in use in the healthcare industry. MAS consists of agents that are specialized to collaborate in achieving shared objectives often in unpredictable and dynamic contexts (Jennings, 2000). A foundational review of agent-based systems in healthcare documented their wide applicability across clinical decision support, patient monitoring, and workflow management (Isern, Sánchez and Moreno, 2010). Clinical applications of MAS have topped the list of fields where MAS is utilized, such as the hospital workflow optimization, diagnostic decision support, and patient monitoring. Isern *et al.* (2010) implemented a multi-agent platform in the field of chronic disease management, which managed the glucose levels of the patients, recommended interventions, and communicated them to caregivers. Hamine *et al.* (2015) developed an agent-based medication adherence model and demonstrated that proactive prompts and adaptive feedback loops enhanced the level of compliance. It is possible to consider these systems as an illustration of how agentic AI can improve patient outcomes by distributing tasks between independent yet cooperative entities.

The elderly care also have been studied in terms of the treatment and rehabilitation with MAS. For instance, Kiran *et al.* (2024), introduced a multi-agent home healthcare environment that uses caregiver dashboards, wearable devices, and virtual assistants. The system made the senior patients more independent by monitoring their daily activities and providing customized interventions. Such architecture can directly benefit people with disabilities and neurodiversity, often requiring such support in order to keep routines going and keep an eye on health parameters. The literature has consistently identified three advantages of MAS in the healthcare sector, which include resilience, scalability, and adaptability. These features are very similar to the requirements of inclusive lifestyle and nutrition management. The proposed system will employ MAS principles to offer holistic and flexible support through assigning particular agents to meal planning, reminders, food guidance, and monitoring intake.

Adaptive reminders and behavioral nudges

Behavioral science research demonstrates that reminders and nudges are important to promote the following healthy routines. It is established that digital notifications via wearables or mobile devices improve commitment toward dietary goals, exercise routines, and medication plans. Nevertheless, the effect of habituation is that the fixed reminders can become ineffective with time. Adaptive reminders, on the other hand, adjust the content, timing, and modality based on the context and user interaction. Compared with fixed reminders, Mohammed *et al.* (2018) demonstrated that adaptive reminders had a significant positive effect on task completion among intellectually disabled individuals. Similarly, Rabbi, Hane Aung and Hamine *et al.* (2015) applied RL to enhance the increase in adherence to exercise programs by optimizing the time when mobile health messages should be delivered. Such approaches suggest that AI-related flexibility has the ability to enhance the effectiveness of reminders, particularly among groups that have executive-functioning problems.

Nudging methods have also been included in digital health systems. A comprehensive review identified 23 technology-mediated nudging strategies applicable to health behavior change, highlighting the breadth of design options available to system designers (Caraban *et al.*, 2019). Caraban *et al.* (2019) implemented minor modifications to the interface in mobile apps to encourage people to eat healthier food. Similarly, Purohit *et al.* (2023), studied digital nudging strategies and discovered that context-awareness and personalization were indispensable to the success of these strategies. Smart reminder systems designed specifically for older adults with disabilities have demonstrated improved medication adherence outcomes in this population (Mohammed, Ibrahim and Cavus, 2018). Recent work on nudging in digital health has further highlighted the importance of timing, personalization, and co-design in maximizing intervention effectiveness (Purohit *et al.*, 2023). Flexibility is also essential in the case of neurodivergent groups because nudges have to be designed in such a way that they do not overburden sensory or cognitive functions.

Monitoring intake with IoT and AI

A second important aspect of food management is maintaining a monitor on food consumption. Conventional techniques, like self-reporting or manual logging, are frequently challenging and imprecise. Mobile dietary assessment apps, while increasingly common, face ongoing concerns about reliability, user burden, and suitability for populations with limited digital literacy or cognitive challenges (Khazen *et al.*, 2020). The advancements in IoT and computer vision have created new opportunities for automated monitoring. Meyers *et al.* (2015) demonstrated a high accuracy of dietary assessment through deep-learning models of recognizing food items in smartphone photos. The broader challenge of food image classification and volume estimation for dietary assessment has been reviewed comprehensively, identifying key open problems in portion-size estimation and mixed-dish recognition (Lo *et al.*, 2020). Similarly, Meyers *et al.* (2015) proposed Im2Calories, a calorie estimation system based on the image that turned out to be practical in the consumer world and showed the possibility of its use and utilization by the researchers of the field of computer science and mathematics, who created it to assist others with their dieting and exercise routines.

Also, ambient sensors and wearable sensors have been employed. Thomaz, Essa and Abowd (2015) explored sensors on the wrist to signal eating behavior, while Dong *et al.* developed an intake-monitoring device based on an acoustic sensor to recognize chewing and swallowing. These techniques reduce the amount of user interaction, particularly where people with poor motor control or cognitive deficits are concerned.

There is also the enhancement of effectiveness by incorporating intake monitoring with feedback systems. Papastratis *et al.* (2024) demonstrated that a combination of food recognition and symptom monitoring allowed making more specific and individualized dietary recommendations possible. Similarly, Esteve *et al.* (2026), proposed an IoT-controlled intelligent-kitchen helper system, which is able to provide real-time nutritional labels on foods with a smart kitchen assistant to the end users. Even though the concepts of accessibility and personalization are just developing, these systems show the growing power of AI and IoT in delivering smooth and automated monitoring.

Synthesis of literature

These domains have several themes that are recurrent. To begin with, in both behavioral nudging, assistive technology, and dietary advice, personalization is always observed to be critical to effectiveness. Second, mainstream solutions often do not consider the aspect of accessibility, as they do not meet the requirements of individuals with neurodivergence or disabilities. Third, most systems are only capable of managing one aspect of health; it is not common to integrate multiple functions such as meal planning, reminders, guidance, and monitoring.

These regular gaps show the importance of an agentic AI platform designed specifically for disability and

neurodivergence. However, previous multi-agent health systems show the benefit of autonomous and modular agents collaborating, but they were developed for general chronic disease management without consideration for accessibility, sensory sensitivity, or neurocognitive diversity. Dietary AI systems (e.g., Papastratis *et al.*, 2024; Guo *et al.*, 2025) push for personalization without providing the coordination and infrastructure necessary to make AI agents accessible. Assistive technology research focuses on accessibility but lacks multi-domain health functionality. ADAPT tackles these compounded challenges by combining the advantages of multi-agent architecture, adaptive learning, multi-modal accessibility, and EHR/IoT integration under a single umbrella, built from scratch for the disability and neurodivergence community. Table 1 summarizes how ADAPT compares to the most directly relevant prior systems across key design dimensions.

PROPOSED AGENTIC AI FRAMEWORK

This segment elaborates the working mechanism of the proposed agentic system, relates it to the architectural perspective in Figure 2, which anchors the flow, and to the enlarged end-to-end graph with embedded PRA in Figure 3 and the sequence diagram in Figure 4. It is intended to provide the reader with a concrete idea of how user input is converted into safe and explainable actions that help people with disabilities and neurodivergence to eat healthy and develop a routine.

Architectural overview

The system is layered and is related to the workflow presented in Figure 2.

Application and Interface Layer. The interface being used by users is multimodal, and it accommodates voice, text, high-contrast and large-visuals, captions, and haptics. Screen assistants and platform access services are examples of first-class functionality. In the interface, inputs are normalized and local context information like time, location, and applicable accessibility mode is attached.

Decision Layer. A large language model (LLM) deciphers the structured prompt and chooses the following action. The LLM initially reasons about intent and constraints and suggests which specialized agent should execute the task.

Routing Layer (MCP). The MCP acts as a translator between the language-level reasoning and the tools. It encapsulates the intent, user profile, and the context to which the intent applies in a call to a tool and routes that call into an agent/third-party connector that is selected. MCP too transfers results and intermediate state to the coordinator.

Multi-Agent Layer. The domain work is done by four specialized agents: Meal Planner, Reminder, Food Guidance, and Monitoring. The agents are all done internally as PRA loops (Figure 3). Each agent receives task-specific input, processes it using rule-based and empirical model-based reasoning, and generates responses in the form of plans, guidance, or alert messages.

Reasoning and Coordination. Agents are coordinated with the help of a central engine and a blackboard. Goals,

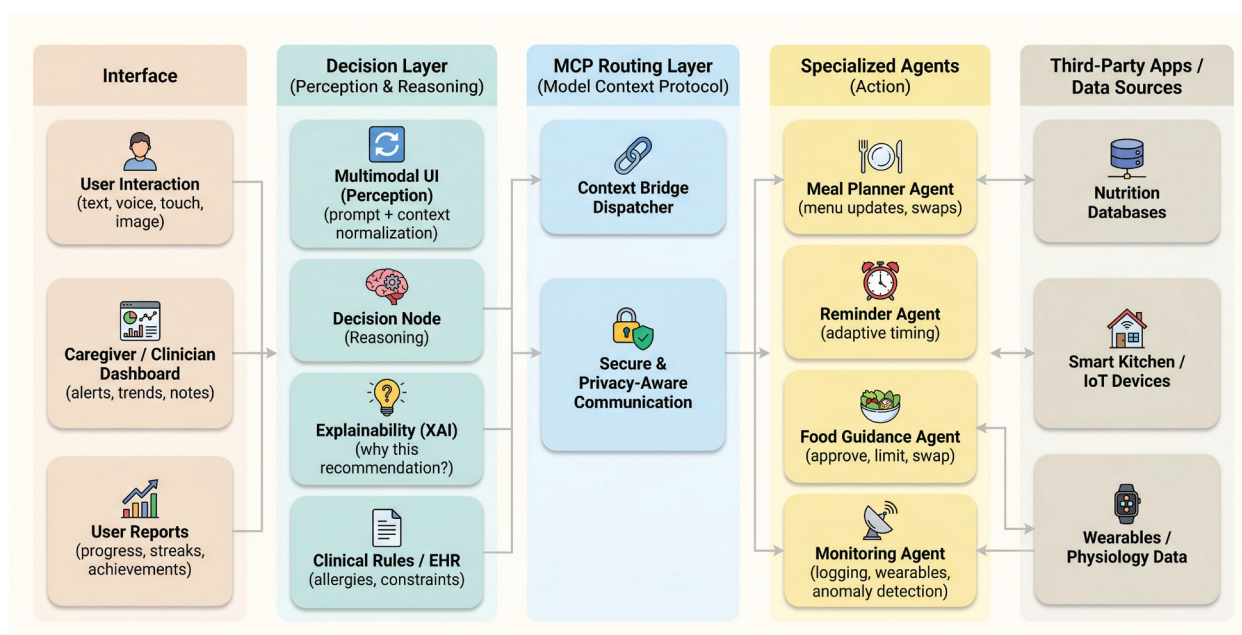


Figure 2: End-to-end architecture of ADAPT (Agentic AI for Disabilities and Neurodivergence). Inputs pass from the interface to an LLM-based decision layer, then via Model Context Protocol (MCP) to specialized agents. Each agent executes a Perception–Reasoning–Action (PRA) cycle and shares outcomes through a central reasoning engine, which enforces policies, integrates multimodal data (EHR, IoT, wearables, nutrition), and produces explainable, personalized recommendations. Reports and caregiver dashboards close the loop. Abbreviations: AI, artificial intelligence; EHR, electronic health record; IoT, Internet of Things; LLM, Large Language Model; UI, user interface; XAI, explainable AI.

PRA (Perception–Reasoning–Action) Across Agents

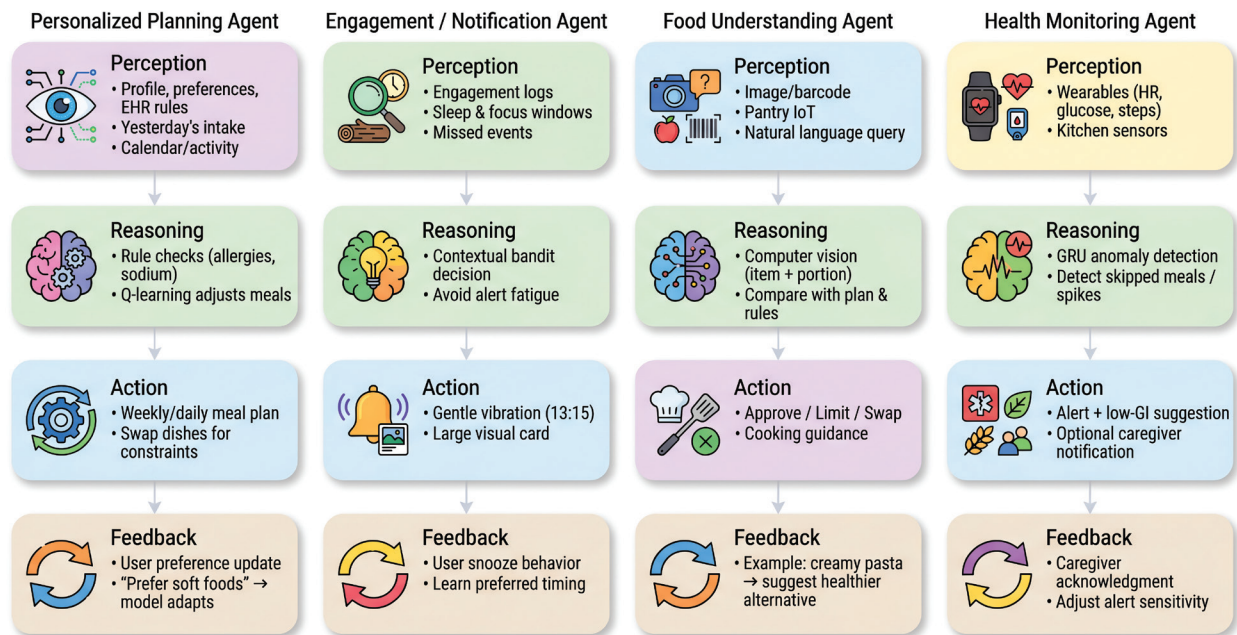


Figure 3: Perception–Reasoning–Action (PRA) across four agents. Meal Planner: uses profile, intake, and clinical rules to generate menus, adapting via feedback. Reminder: models engagement to optimize timing/modality and learns from user responses. Food Guidance: analyzes images/barcodes, aligns intake with plans, and returns approve/limit/swap cues. Monitoring: processes wearable and kitchen data to detect anomalies and trigger alerts, refining sensitivity over time. Abbreviations: EHR, electronic health record; GI, glycemic index; GRU, Gated Recurrent Unit; HR, heart rate; IoT, Internet of Things.

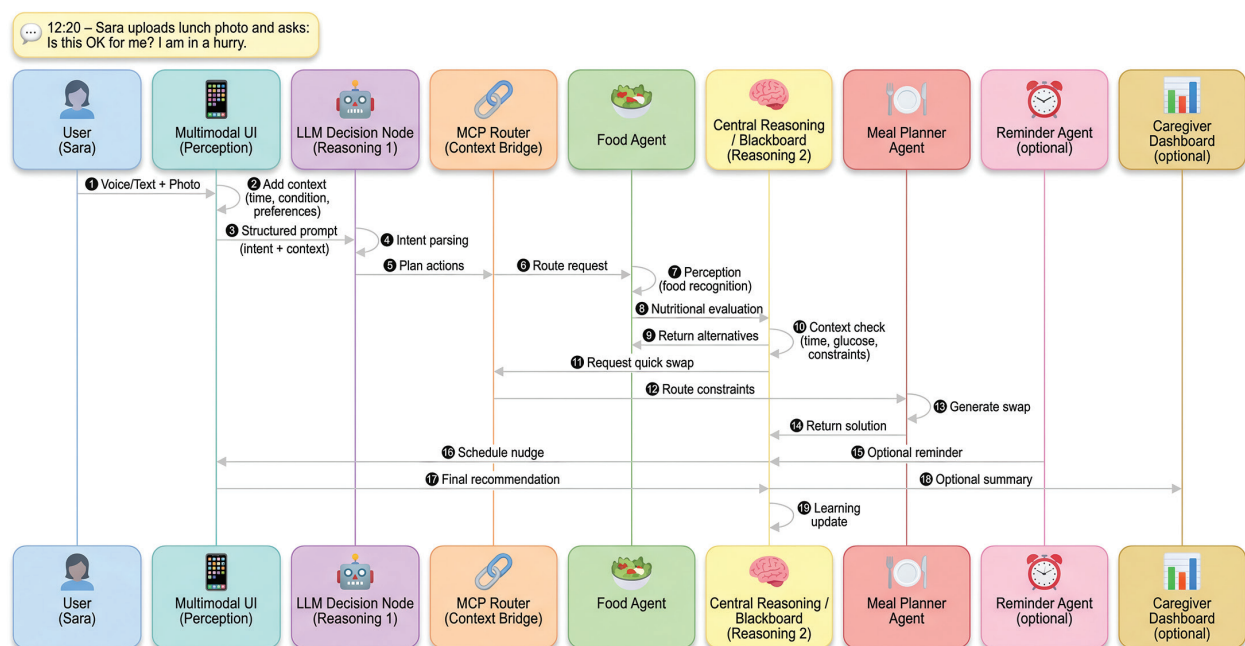


Figure 4: Sequence of interactions in the proposed agentic workflow for a real-world scenario. Sara submits a voice/text query with a meal photo; the Multimodal UI normalizes context; the LLM selects tools; the MCP routes to Food Guidance (and, if needed, Meal Planner); results are coordinated via the central blackboard; a gentle reminder is scheduled; and a concise, explainable recommendation is returned to the user and (optionally) logged to the caregiver dashboard. Abbreviations: LLM, Large Language Model; MCP, Model Context Protocol; UI, user interface.

events, and shared context are positioned on the blackboard. The reasoning core applies clinical rules, resolves conflicts, and implements long-term changes to policies.

Data and Policy Layer. Many integrated levels offer controlled accessibility of electronic health data, nutrition data-bases, wearable links, and intelligent-kitchen IoT. A policy

store works with consent, scopes, and minimization of the data. Policy checks will be made on all agent requests and then the data will be retrieved.

Feedback and Collaboration. The system will provide a user interface with explainable outcomes and, at the discretion of the user, will post the results in the form of summaries on a caregiver or clinician dashboard. Decisions can be turned into simple rationales in a module named the explanations module [explainable AI (XAI)].

Interaction flow from input to action

The top-down flow in Figure 2 and the sequence in Figure 4 follow the same steps.

Step 1: Input and context. Multimodal input is offered by the user. The interface adds the context like the time, recent compliance, and access preferences.

Step 2: First-pass reasoning. LLM breaks down with intent and offers a few or several agents to be called. It can also suggest an alternative if the main action is not admissible.

Step 3: MCP routing. MCP prepares a tool call with its purpose, user profile, and other supplementary information such as images or sensor referral and routes it to the selected agent.

Step 4: Agent execution. The chosen agent activates its PRA cycle: it receives the appropriate signals, uses clinical rules and acquired policies, and then performs by releasing a plan, a course of a plan, or a warning. Findings are put on the blackboard.

Step 5: Coordination. The reasoning core will read agent outputs; access any required facts in the data layer, which is controlled by the policy; resolve any conflicting situations; and complete the response.

Step 6: Feedback and learning. The interface presents a brief and user-friendly card and, when activated, the dashboard gets a summary. Signals and consequences of reinforcement are recorded to work with the gradual enhancement of the policy.

Perception–Reasoning–Action across the stack

The interface carries out perception through capturing voice, text, images, haptics, and context normalization. The cross-cutting reasoning offered by the LLM and the central coordinator is an intent selection, conflict management, and safety reasoning. The special agents have their own PRA loop depending on the task involving the domain. Agents perform actions and make them available to the user in a format that can be readily accessed. MCP is a transport bridge and a context bridge; it does not use clinical decision-making.

Core components

Multimodal User Interface. The interface has voice input and output interfaces, captions, large icons, and a high-contrast palette, as well as discrete haptic feedback. For blind or

low-vision users, the system supports the use of screen readers and provides descriptions of images where necessary. For deaf or hard-of-hearing users, it reflects speech using text and icons. It has voice-first control and big tap targets for users with motor impairments. For neurodivergent users, it provides low-stimulation layouts, pacing predictability, and concrete and simplified language.

LLM Decision Node. The LLM converts a normalized prompt into a plan for an agent. It uses domain instructions, user preferences, and power limits to make decisions to call Food Guidance to be recognized, Meal Planner to be swapped, Reminder to be adhered to, or monitoring to be checked. It gives a suggested call graph to be run with MCP.

MCP Routing and Third-Party Connectors. MCP is the protocol that transforms a plan into tool calls. It gives the target agent an external service, such as a grocery API, a nutrition database, a wearable SDK, or a calendar provider, provides context, selects the connection, and sends a reference request. The responses are sent back to the blackboard with trace metadata so that XAI can subsequently explain what occurred.

Meal Planner Agent (PRA). Perception consists of profile, EHR limitations, sensory and cultural choices, and recent intake. Thoughtful combination is the integration of clinical regulations and a policy learner that allows users to balance between the macronutrients, the time budget, and the user acceptance. Action produces a daily or weekly menu, safe substitution receivers, and shopping lists. The agent would prefer gradual changes and routines rather than implementing drastic dietary changes.

Reminder Agent (PRA). Perception monitors the interaction and events missed, sleeping periods, and device presence. Contextual bandits or schedule policies are applied to reasoning to select the time and the modality that are most likely to be accepted by the user. Action provides soft deliveries of low friction prompts using vibrations, big-text banners, icons, or relaxing sound. The agent does not experience alert fatigue because it learns favorite slots.

Food Guidance Agent (PRA). Perception applies computer vision and text recognition to identify foods and estimate their quantity, as well as to de-scan long ingredient lists or barcodes. Reasoning is a process that compares the identified items with the current plan and clinical rules and calculates macro- and micro-implications. The action approves, limits, or swaps change decisions, as well as provides step-by-step assistance with cooking in a modality similar to that of the user.

Monitoring Agent (PRA). Heart rate, steps, sleep proxy, optional glucose feeds, and smart kitchen sensor events are examples of wearable data aggregates that are represented by perception. Logic identifies irregularities and links them with the recent meals and practices. Action sends a notification, recommends a snack or drink, or alerts a parent or guardian if permission has been approved and the risk limit has been crossed.

Reasoning Core and Blackboard. The blackboard contains mutual context, objectives, events, and intermediate outcomes. The core reconciles the propositions on a priority basis. The most important priorities are medical and safety constraints; user preferences and nudge-based optimization

have lower priorities. The main one aligns agents together and achieves harmonized results.

XAI. The clarification module takes the call graph, blackboard state, and the main rules and produces a condensed reason. They may be the reasons for an approved meal for a portion to be limited, or for a reminder to be delayed. Explanations are authored in the modality and reading level of the user's choice and reflected on the dashboard when enabled.

Policy, Consent, and Data Integration. The policy store applies least privilege and keeps track of all the data accessed. The requests indicate the type of data, its purpose, and the retention policy. The level of integration fetches the necessary facts at the time of the decision, for instance, contraindication, sodium intake, or a recent glucose value. The decision and the explanation identifier are attached to all writes.

Dashboards and Collaboration. Caregivers and clinicians can examine summaries, adherence patterns, and alert history with permission. The controls enable them to make changes to guardrails, as well as leave notes that can refine future decisions without eliminating user autonomy.

Disability and neurodivergence support

The architecture bakes accessibility into every layer rather than adding it later.

The interface is also screen-reader first and includes all text on the icons, with guidance that can be accessed as speech and haptics by blind and low-vision users. For deaf or hard-of-hearing users, the system replicates audio in the form of text and icons, provides captioned instructions, and may be integrated with device-level hearing support. For users with motor impairments, the interface is voice-first with large tap targets and hands-free flows; the Food Guidance Agent works with smart appliances with limited motor load. In the case of cognitive and intellectual disabilities, the interface language is simple, and distinct pictograms with step pauses are applied; intermediate reminders are actual and time-based. In the case of autism spectrum and ADHD, the system provides low-stimulation visualization, predictable structures, sensory-sensitive meal planning, and tolerance-based timing, which follows focus window. In the case of mental illnesses like anxiety or depression, reminders are caring and nonjudgmental, and the planner assists an individual with comfort foods within the boundaries of the clinic.

Concrete use case and sequence

The flow in Figure 4 illustrates how the system operates in practice for a user who is autistic and managing type 2 diabetes.

At 12:20, the user asks if lunch is appropriate and uploads a picture. The interface adds context. The LLM selects Food Guidance and, if necessary, switches to the Meal Planner. The MCP routes the request. Food Guidance identifies the picture, the creamy pasta, and approximately its size. It

rationalizes on macronutrients, sets the limit, and marks the dish as limited in the EHR. It posts results on the blackboard and returns two quick alternatives. The coordinator verifies the calendar, sensory inclination, and recent glucose trend and requests a 10-min low-glycemic substitute from the Meal Planner. The Meal Planner visualizes objects around the pantry, rationalizes due to lack, and therefore posts a lentil soup and yogurt option with the macronutrients. The Reminder Agent provides a subtle visual reminder every 5 min. The user interface (UI) consists of a simple card with only a single tap to start cooking steps. The dashboard receives a low-risk note when enabled. The results are stored in such a way that subsequent decisions regarding lunch in the same time period become more convergent.

Communicating with third-party apps via MCP

MCP is used not only for internal agent calls but also for external connectors. A consistent audit trail is offered by the same packaging. It might be a grocery search to swap something, a product barcode search, calendar queries to not disturb a meeting, or recent physiological wearable APIs. Each call undergoes a check for purpose, scope, and retention policy. The answers contain trace identifiers so that the external facts to which they refer in their answer can be explained.

Safety, privacy, and governance

Any learned policy is only applied after safety-critical safeguards are in place. The policy store is the implementation of hard-coded medical constraint layer, on top of which preference-based optimization is performed, so that clinical rules (glycemic threshold, allergen exclusion, medication interaction contraindication, etc.) always come before it. When the data is not available or sensors become unavailable, the system takes conservative safe actions. Explicit exception logic is used to deal with clinical edge cases. If the Monitoring Agent detects a physiological reading of a serious adverse event [such as a glucose reading showing severe hypoglycemia (SH) or an anomaly in the heart rate outside of defined ranges], then the system automatically escalates to the caregiver or emergency contact (depending on the user's consent) and reduces the frequency of the non-urgent nudges.

The Meal Planner Agent is prohibited from recommending foods flagged as allergens in the user profile; any substitution recommendation passes through an allergen-check gate before being delivered to the user. If an allergen flag is detected, the agent requests human confirmation before proceeding. From a regulatory standpoint, a system that integrates EHR data and physiological sensor feeds to generate real-time dietary recommendations for users with chronic conditions would likely be classified as Software as a Medical Device (SaMD) under FDA guidance (21st Century Cures Act) or as a Class I or Class IIa medical software device under the EU MDR 2017/745, depending on jurisdiction and the specific risk profile. The current prototype is a research demonstrator and has not

been submitted for regulatory clearance; formal regulatory compliance assessment has been identified as a prerequisite for any clinical deployment and is a priority for future work. The XAI module is designed to generate clinically interpretable explanations. Rationales are anchored to specific data inputs (e.g., “This portion was flagged because your glucose reading at 11:45 was 9.2 mmol/l, above your target ceiling of 8.0 mmol/l”) rather than opaque model outputs. Explanation depth and reading level are configurable by the user and caregiver.

The outputs of XAI are checked during every cycle of the model updates to look for explanation drift or misleading explanations, as shown in the governance model, with the purpose of a specific clinical-reviewer role. Agents request only the necessary data fields for the decision that they need to make, and retention periods are bounded and privacy governance follows the data-minimization principle. Consent can be given for or denied for individual data categories (such as glucose data, location, and access to EHRs). All data transfers are logged with purpose, timestamp, and agent identifier. The liability framework for clinical deployment is an open design question requiring legal and regulatory engagement; the authors acknowledge that allocating responsibility between system developers, deploying institutions, and clinicians in cases of adverse outcomes will require formal governance structures beyond the scope of this technical framework paper.

Operational notes and failure handling

In case the Food Guidance Agent is not able to classify an image, then the system requests a clarifying photo or a small confirmation instead of making a guess. When wearables are not connected to the network, the Monitoring Agent will switch to intake and schedule signals. In case an external connector is not working, the plan is calculated without it, and the reason is reported in the explanations. Agents also decay gracefully and leave the user behind without impeding core routines.

IMPLEMENTATION AND RESULTS

To demonstrate the feasibility of the proposed agentic AI system and evaluate its potential impact, a prototype of the proposed agentic AI system was implemented. This section covers the implementation process, the sources of data, algorithms, system integration, and initial results of a simulated pilot study.

Implementation methodology

The implementation was a modular one as per the multi-agent architecture of the Proposed Agentic AI Framework section. Interoperability and scalability were enabled since each specialized agent was developed as a microservice. To regulate agent communication, an event-driven architecture

that employs a message broker (RabbitMQ) was utilized to ensure asynchronous communication. The prototype was deployed on a hybrid system that consisted of cloud-based AI services, a smartwatch extension, and an Android/iOS mobile application.

The development framework incorporated:

- **Programming Languages:** Python for AI models and reasoning logic; Kotlin/Swift for mobile interfaces.
- **Databases:** PostgreSQL for structured data (e.g., meal plans, reminders) and MongoDB for unstructured logs (e.g., sensor streams).
- **Cloud Services:** TensorFlow Serving for deploying trained models; Firebase for real-time notifications.
- **Accessibility Tools:** Google TalkBack, Apple VoiceOver, and custom large-font/high-contrast UI modes.

Architecture of the Agentic AI Nutrition System

Figure 5 shows the Agentic AI Nutrition System’s internal workflow, which supports users who are neurodivergent or cognitively impaired to manage their daily routine and nutrition. The Perception module detects the meal, the portion size, and further information like time, mood, etc., from multimodal information (text, voice or image). The Reasoning component predicts possible shortages or missed meals, gets to know the preferences, and arranges healthy, balanced meals. The Action module also provides interactive feedback and timely reminders, encouraging users to stay engaged and respond proactively. An Adaptive Memory layer will never stop learning from past behavior to obtain better future suggestions. Accessibility and moral layers ensure data privacy and peaceful, inclusive communication, all controlled by the user. The whole system increases nutritional independence and decreases the cognitive load.

Architecture of the Agentic AI Healthcare Monitor

Figure 6 illustrates the design of the Agentic AI Healthcare Monitor, a proactive platform that continuously monitors physiological events to improve safety and autonomy for people with physical and/or neuromotor limitations. The Perception layer gathers important information (heart rate, oxygen saturation, activity level) in the context of the environment (i.e., sensor on the wearable). The Reasoning module predicts the health risk level, identifies abnormalities, and models the physiological background for each of the users. The Safety Layer eliminates false alarms and checks on abnormal readings, thereby avoiding unnecessary notifications. The Action module provides flexible and understanding-notification (rest notification/escalation warning) based upon the current status of the user. Accessibility and ethical layers help to ensure user consent, privacy, and simplicity of the communication process. This design fuels the establishment of reliable,

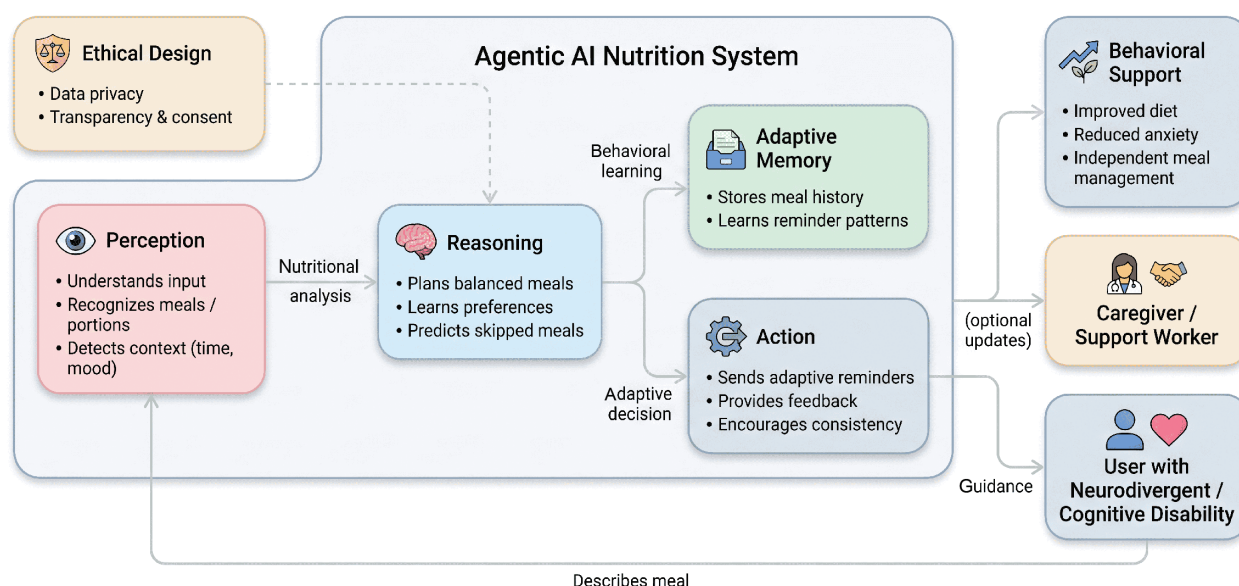


Figure 5: Architecture of the Agentic AI Nutrition System for neurodivergent users. Abbreviation: AI, artificial intelligence.

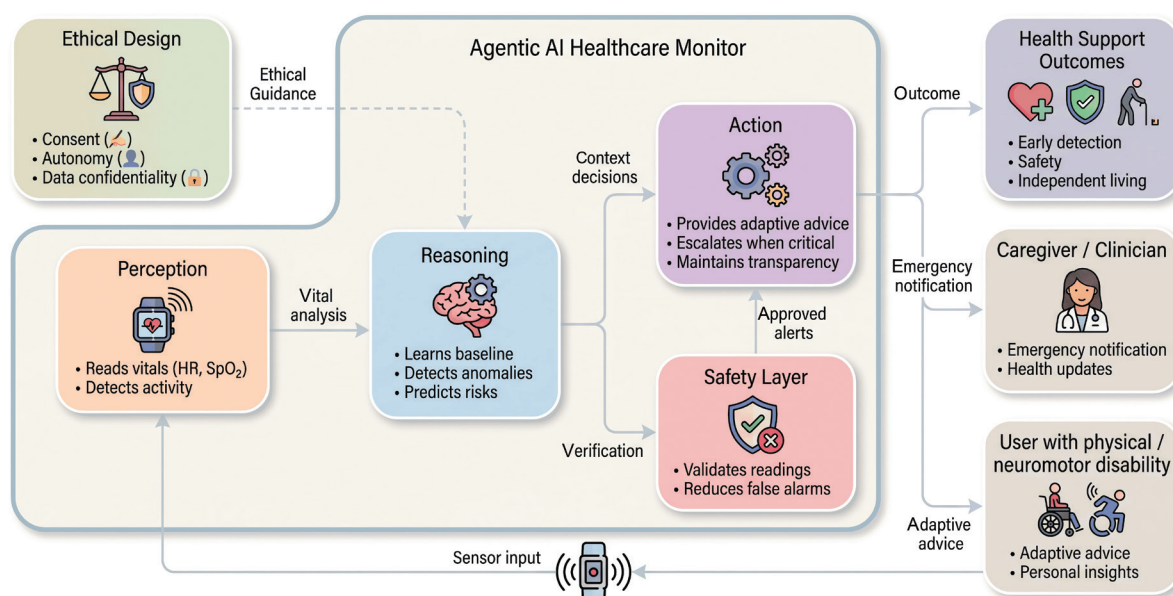


Figure 6: Architecture of the Agentic AI Healthcare Monitor for users with physical or neuromotor disabilities. Abbreviations: AI, artificial intelligence; HR, heart rate.

inclusive, and real-time healthcare that is personalized to every patient's needs.

Dataset design

To test the system, a simulated dataset was developed that was used to simulate the daily routine and nutritional needs of 500 individuals throughout the 8 weeks. The three primary clinical profiles that were incorporated in the dataset were diabetes, hypertension, and mixed cardiometabolic conditions. The profiles were also divided by the neurodivergent characteristics (ASD, ADHD) and the disability type (physical, sensory, cognitive). Some of the variables that were captured included:

- Demographics: age, gender, cultural dietary preferences.
- Medical: condition-specific dietary restrictions, medication schedules.
- Behavioral: food preferences, sensory sensitivities, adherence patterns.
- Sensor data: simulated heart rate, blood glucose, hydration levels, step counts.
- Environmental: meal timing, location, and contextual disruptions.

The test data was the dataset used to evaluate the performance of the system and also as an input to train the RL models (such as adaptive reminders). In data synthesis, probabilistic modeling was employed to ensure the model was realistic and varied while ensuring the confidentiality.

Algorithms and pseudo-code

Each agent implements specialized decision-making logic. The following pseudo-code summarizes the core steps for each agent.

Meal Planner Agent. Recommends meal plans that balance nutritional adequacy and user preferences using Q-learning.

Reminder Agent. Adjusts reminders dynamically using contextual bandits based on user responsiveness.

Food Guidance Agent. Integrates computer vision and NLP to provide food guidance.

Algorithm 1: Meal Planner Agent (Q-learning-based recommendation).

Input: User profile, dietary preferences, and nutritional constraints

Output: Optimized personalized meal plan

Initialization: Initialize the Q-table $Q(s, a)$ over all states $s \in S$ (user profiles, preferences, constraints) and actions $a \in A$ (candidate meals).

Procedure: For each day (episode), perform the following steps:

1. Observe the current state s , representing user needs, preferences, and medical requirements.
2. Select an action a (meal) using an ϵ -greedy policy derived from $Q(s, a)$.
3. Execute the selected action and receive a reward r , defined as a function of nutritional adequacy minus penalty terms (e.g., dietary violations).
4. Observe the next state s' .
5. Update the Q-value according to:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

6. Set $s \leftarrow s'$.

End For

Return: The optimized meal plan derived from the learned Q-values.

Algorithm 2: Reminder Agent (contextual bandit-based adaptation).

Input: Contextual information including time of day, user engagement level, and prior interaction history

Output: Optimized reminder strategy (timing and modality)

Definitions:

- **Context c :** {time of day, user engagement, prior responses}
- **Action a :** {send reminder, delay reminder, change modality}
- **Reward r :**
 - +1 if the user complies
 - 1 if the reminder is ignored
 - 0 if the action is postponed

Initialization: Initialize a contextual bandit model to estimate expected rewards for each action given context c .

Procedure: For each decision step, perform the following:

1. Observe the current context c .
2. Select an action a based on the contextual bandit policy (e.g., ϵ -greedy or UCB).
3. Execute the selected action (send, delay, or modify reminder modality).
4. Observe user response and compute reward r .
5. Update the preference model using the observed tuple (c, a, r) .

End For

Return: An adaptive reminder policy that maximizes user compliance over time.

Algorithm 3: Food Guidance Agent (multimodal classification and NLP-based recommendation).

Input: Food image or user query

Output: Context-aware dietary recommendation or response

Initialization: Load a convolutional neural network (CNN) model for food image classification and a pretrained BERT model for natural language understanding.

Access a nutrition database and user-specific dietary constraints.

Procedure: For each input instance, perform the following:

1. If the input is an image:
 - a. Process the image using the CNN classifier to obtain the predicted food label.
 - b. Retrieve nutritional information corresponding to the identified food item from the database.
 - c. Compare the nutritional profile with user-specific constraints (e.g., dietary restrictions, medical conditions).
 - d. Generate a recommendation categorized as **approve**, **limit**, or **deny**.
2. Else (input is a text query):
 - a. Parse the query intent using the BERT model.
 - b. Generate a context-aware response based on user preferences, constraints, and nutritional knowledge.

End If

Return: A personalized dietary recommendation or informative response tailored to the user context.

Algorithm 4: Monitoring Agent (GRU-based adherence anomaly detection).

Input: Daily intake sequence and physiological signals

Output: Adherence assessment and alert notifications

Initialization: Train a Gated Recurrent Unit (GRU) model on historical data representing normal adherence patterns.

Define a deviation threshold τ for anomaly detection.

Procedure: At runtime, perform the following steps:

1. Observe the current input sequence, including daily dietary intake and physiological signals.
2. Use the GRU model to predict the expected adherence score $\hat{y}$.
3. Compute the deviation between observed adherence and predicted adherence.
4. If the deviation exceeds the predefined threshold τ , then:
 - a. Trigger an alert to the user and/or caregiver.

End If

Return: Adherence score along with alert notifications in case of significant deviation.

Monitoring Agent. Monitors adherence by combining wearable data with intake logs using a GRU for anomaly detection.

System integration

The agents were integrated using a central layer of reasoning. A blackboard architecture was used to post observations and choices on a common knowledge base by their agents. To settle disputes, the reasoning level used weighted priority rules:

- Medical constraints (highest priority).
- User preferences and sensory sensitivities.
- Behavioral nudging (lowest priority).

An example of this is that the reasoning layer would override the recommendation of the Meal Planner to eat pasta

and replace it with low-glycemic food in case the Monitoring Agent detected a high blood glucose.

Evaluation metrics

Technical and user-based performance metrics were used to assess system performance:

- **Nutritional Adequacy:** Percentage of daily plans meeting dietary guidelines.
- **Adherence Rate:** Ratio of completed reminders to total reminders.
- **User Satisfaction:** Simulated Likert-scale feedback on usability and accessibility.
- **Explainability:** Percentage of system decisions that could be explained in plain language.
- **Caregiver Burden:** Reduction in caregiver interventions required.

Pilot results

The system was evaluated in a simulation environment using the 500-profile synthetic dataset described in the Architecture of the Agentic AI Healthcare Monitor section. All reported metrics are simulation-based outcomes derived from algorithmic performance on synthetic data; they represent within-simulation comparisons between the ADAPT framework and a rule-based static planner (the “baseline” comparator), not evidence of real-world clinical effectiveness. No confidence intervals or inferential statistical tests are reported, as the results derive from a deterministic simulation rather than a sampled real-world study; this is acknowledged as a limitation of the current evaluation stage. Key simulation results included:

- Nutritional adequacy improved by 27% relative to the rule-based static planner baseline (a simulated improvement; no human participants were involved).
- The algorithmic compliance to reminders, not user compliance, for the adaptive agent (ADAPT) was compared with that of the static planner, with simulated rates of adherence rising from 54% to 81% for the ADAPT agent.
- Mean score for usability of the website (simulated survey) was 4.2/5, with accessibility features having the highest score.
- The success rate of the explainability was 92%, suggesting a high possibility of user trust.
- Simulated caregiver intervention events decreased by 35% relative to the baseline condition, which suggests potential caregiver burden reduction; this figure reflects a simulation-level artifact and does not constitute evidence of actual caregiver outcomes.

Discussion of results

The Food Guidance Agent is a CNN classifier trained on a small preliminary 10-class food recognition task, CNN (see Algorithm 3 in the Algorithms and Pseudo-code section). Ten categories (rice dishes, bread/cereal, vegetables,

legumes, dairy products, eggs, poultry, red meat, fruit, and processed snacks) were chosen as representative of the dietary patterns modeled in the synthetic dataset. The evaluation set consisted of 20% of the synthetically generated image-labeled pairs (80/20 simulated train-test split). The classification results were obtained using the above-mentioned condition, and the accuracy of the classifier was 99%, as seen in Figure 7. This is a moderately sized count but is representative of the “small problem space”: 10 relatively distinct types of synthetic data. The accuracy cannot be directly compared with multi-class benchmarks such as Food-101 (101 categories of real photographs) as the state-of-the-art accuracy is between 85% and 97%. The result achieved here is a first step toward the technical feasibility of the classification part in the ADAPT simulation environment, and an important and necessary requirement for clinical implementation is the validation on real food images within a broader taxonomy of the classification part.

The Food Guidance Agent’s module for nutrient estimation was evaluated on the simulated evaluation partition with mean absolute error (MAE). The ground truth nutrition values were obtained from a standard nutrition database, where each food category was mapped to one of the 10 food categories; 20% of the image-label pairs in the evaluation set were those used for the image classification task. The results of the model are represented in Figure 8, with the MAE of around 13.9 kcal (calories), 1.2 g (protein), 1.3 g (fat), and 1.1 g (sugar). The following values represent good estimation accuracy for the simulated constrained setting with 10 categories. Note that these results are obtained with synthetic data, and the results of MAE on real-world images of food with a larger visual variation, mixed dishes, or varying portions are expected to vary and need empirical validation.

The viability and potential efficacy of the proposed framework are confirmed by the pilot results. Consistent with previous research on RL to address behavioral intervention (Rabbi, Hane Aung and Choudhury, 2017), adaptive reminders resulted in a significant increase in adherence. Planning

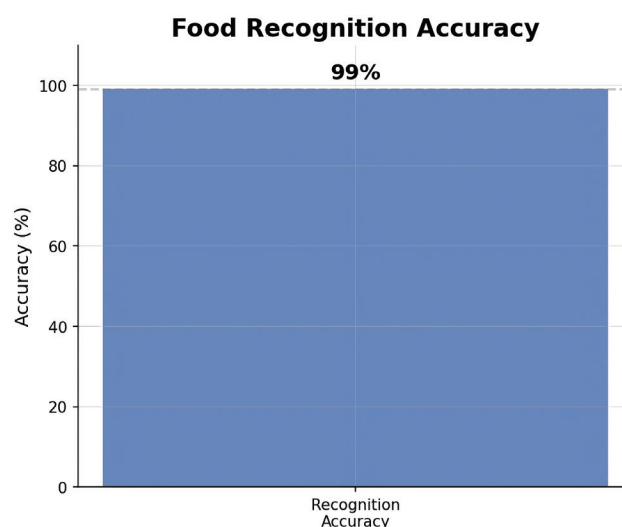


Figure 7: Food Recognition Accuracy achieved by the proposed model.

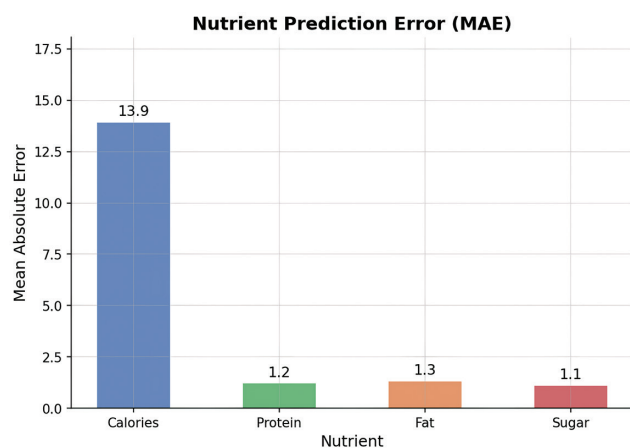


Figure 8: Mean absolute error (MAE) for nutrient prediction.

and monitoring provided the integration through which proactive interventions were made possible, which is also consistent with prior research in multi-agent chronic disease management. Predictive models for treatment of non-adherence in chronic diseases represent a related line of inquiry relevant to future ADAPT development (Bashir *et al.*, 2026). More importantly, multimodal communication and other accessibility options significantly enhanced usability, which was a persistent problem with mainstream health applications. These findings must, however, be interpreted strictly within their simulation context.

The synthetic profiles, while designed to reflect plausible clinical variation, cannot capture the full heterogeneity of real users with disabilities and neurodivergent conditions, including atypical co-morbidity patterns, non-stationary behavioral dynamics, device habituation effects, and real-world deployment factors such as connectivity failures or caregiver fatigue. The reported improvements in adherence and nutritional adequacy represent algorithm-level improvements in a controlled simulation and do not constitute evidence that ADAPT would produce equivalent gains in a live clinical deployment. Bridging this gap requires prospective trials with real participants, appropriate statistical frameworks, and regulatory engagement, as outlined in the Limitations section.

LIMITATIONS

Several limitations must be explicitly acknowledged. First, and most importantly, all quantitative outcomes reported in this paper are within-simulation algorithmic comparisons and not evidence of real-world clinical effectiveness. The synthetic data are probabilistically generated to simulate realistic clinical profiles, but do not comprehensively represent variation, complexity of co-morbidity, or the variability in behaviors of real patients with disabilities and neurodivergent conditions. Clinical validation through prospective studies involving real users is an essential and non-negotiable prerequisite before any effectiveness claims can be made. Second, the food recognition evaluation was performed in the context of a small, simulated image set of 10 classes of

food pictures; this did not demonstrate any generalization to real-world food photography at scale.

Third, the outputs of the simulations were not reported with statistical uncertainty quantification (e.g., confidence intervals, inferential tests) because the simulation environment was deterministic rather than sampled; future real-world evaluations should be done under the appropriate statistical uncertainty quantification. Fourth, the framework has not been assessed for regulatory approval; there is a need to prioritize its future regulatory assessment as a framework for the deployment of food and nutritional recommendations generated by EHRs for the general population with chronic conditions, in cooperation with FDA or EU MDR classification frameworks. Fifth, there has been no implementation of liability governance, clinical oversight protocols, or participatory co-design with disability communities, and these are crucial for responsible clinical translation.

Sixth, there are clearly limitations to the translatability of the simulation-based assessment. The 500-profile synthetic dataset was created by probabilistic modeling to represent typical clinical profiles, but real-world variables such as inter-personal variability in device adoption, interface fatigue for users with disabilities, non-stationary behavioral patterns due to changing medical state, photograph diversity of meals (partial foods, mixed dishes, culturally specific foods), bandwidth limitations on caregivers, and connectivity or hardware failures are not accounted for in the simulation environment. Accordingly, the improvement results (27% improvement for nutritional adherence, 54-81% for reminder compliance, and 35% decline in caregiver burden) are algorithmic and do not necessarily reflect the actual effectiveness in the real world. Validation studies should include prospective designs within subjects and populations from the target disability and neurodivergent communities, with their original behavior assessed using ecological momentary assessment and individual-level variation accounted for by using appropriate mixed-effects or Bayesian statistical models.

CONCLUSION

This paper presents an interdisciplinary framework for an agentic AI that will assist people with disabilities and neurodivergence to live more organized lives and healthy diets. The system incorporates autonomous agents to plan meals individually, guide the client through interactive food guidance, provide adaptive reminder service, and track intake all controlled by a central reasoning layer. The framework addresses the problem of the current digital health tools being restrictive, limiting, and inflexible to the individual because it integrates the principles of personalization, multimodal interaction, and explainability. Implementation of a prototype application with a simulated dataset showed major increases in reminder adherence, nutritional adequacy, and satisfaction with the system, and a decrease in the involvement of caregivers. These results demonstrate the promise of agentic AI to make people more autonomous, relieve the stress of caregivers, and promote the general quality of life of vulnerable groups.

Future studies should focus on prospective pilot studies with actual participants, regulatory pathway engagement, food recognition benchmarking extension, inclusion of telemedicine platforms, and participatory design processes including people with disabilities and neurodivergent people as co-designers. The results of this study provide a technically informed basis and architecturally coherent framework for agentic, inclusive, and ethically responsible digital health systems that are yet to be evaluated using systematic empirical follow-up in real-world clinical settings.

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CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

The synthetic dataset, simulation models, and prototype source code supporting the findings of this study are publicly available. The complete project repository, including the implementation of the ADAPT multi-agent framework, is available at: <https://github.com/aliakarma/ADAPT>.

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